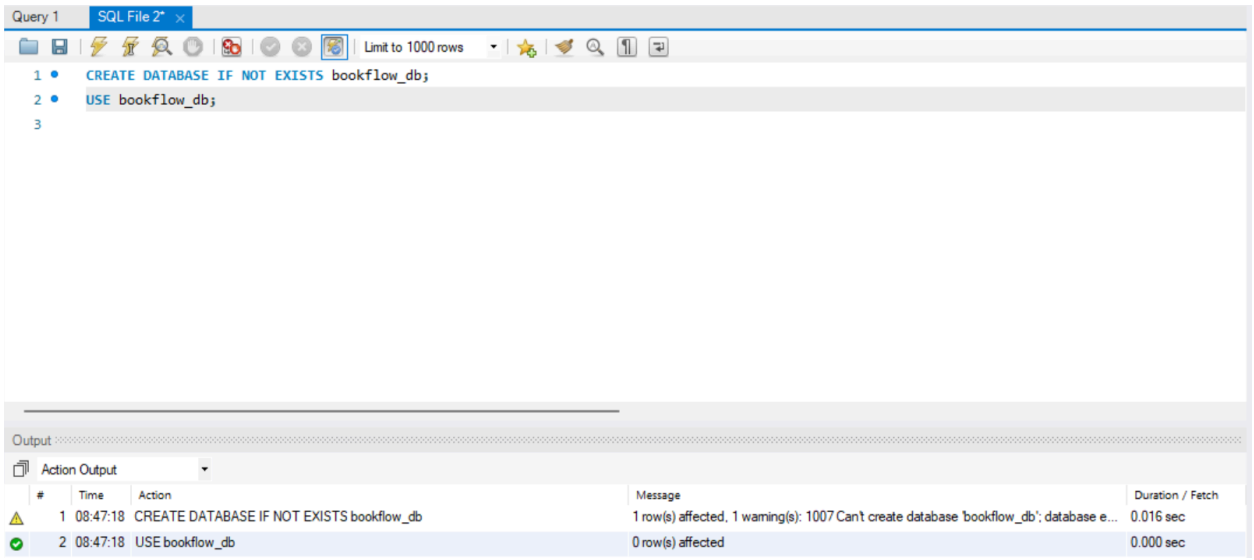
Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

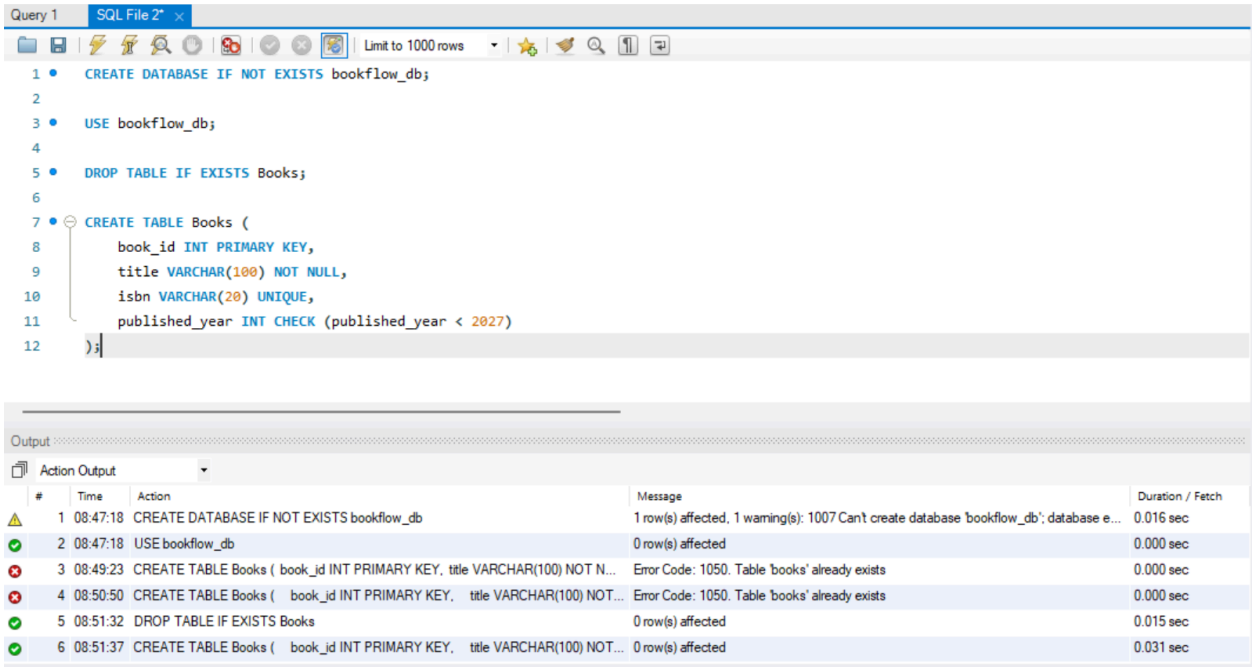
Week-2

Name : D. Rohini Tanmayi
Roll Number : 2520030349

Step 1 — Create Database



Step 2 Table Creation (DDL) with Constraints :



Step 3 & 4 — Insert Books and Display

```
14 •  INSERT INTO Books (book_id, title, isbn, published_year) VALUES
15    (1, 'The Great Gatsby', '9780743273565', 1925),
16    (2, 'To Kill a Mockingbird', '9780061120084', 1960),
17    (3, '1984', '9780451524935', 1949);
18 •  SELECT * FROM Books;
19
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

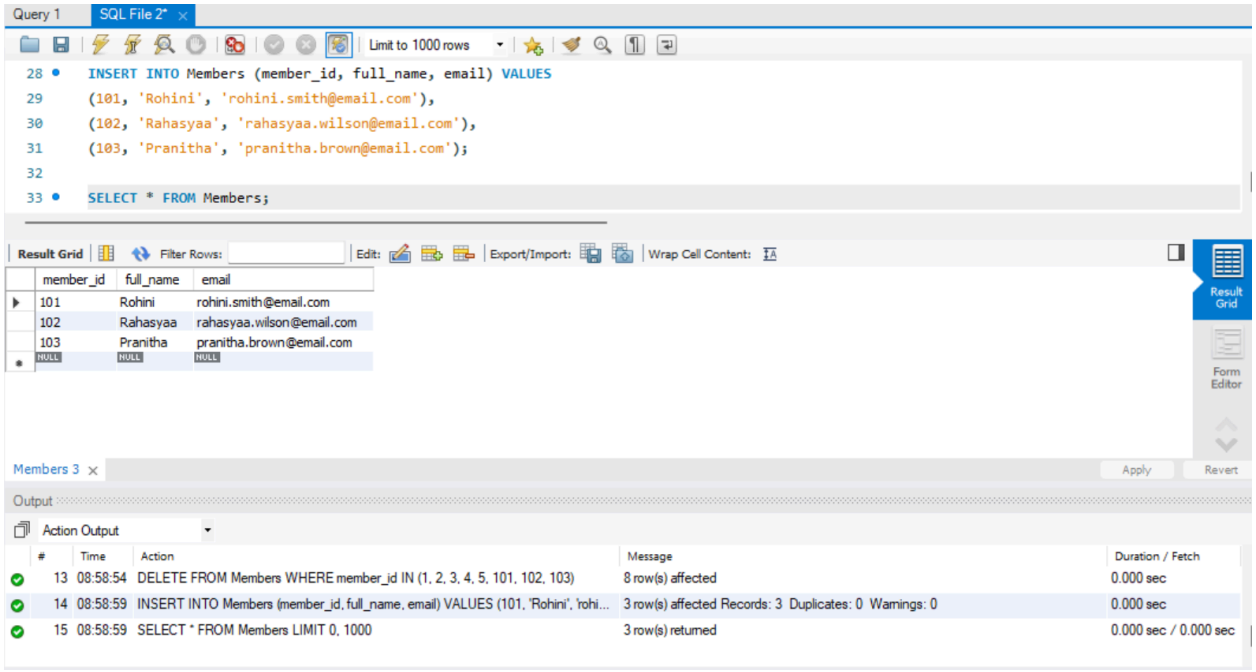
Books 1 | Apply | Revert

Output

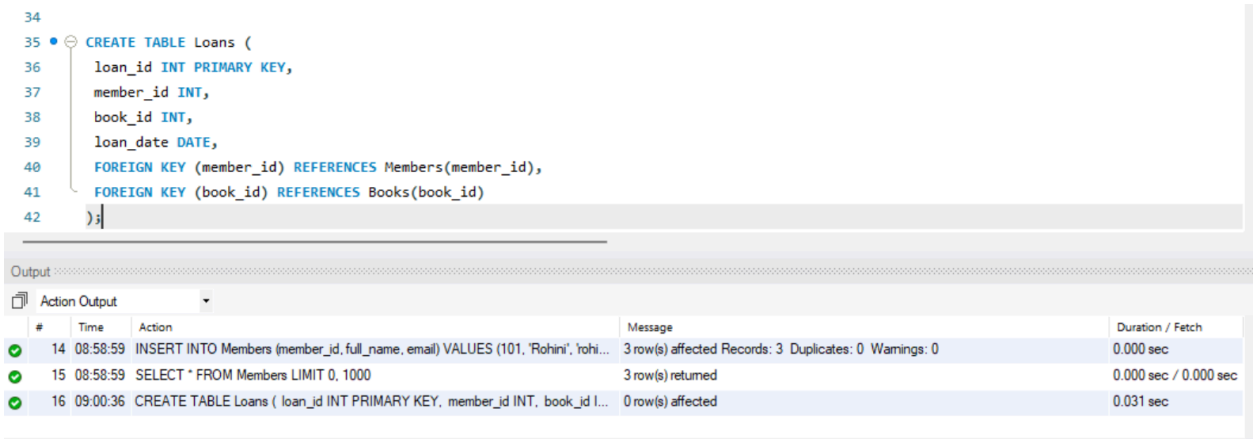
Action Output

#	Time	Action	Message	Duration / Fetch
3	08:49:23	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT ...	Error Code: 1050. Table 'books' already exists	0.000 sec
4	08:50:50	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) N...	Error Code: 1050. Table 'books' already exists	0.000 sec
5	08:51:32	DROP TABLE IF EXISTS Books	0 row(s) affected	0.015 sec
6	08:51:37	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) N...	0 row(s) affected	0.031 sec
7	08:52:58	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great ...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
8	08:52:58	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

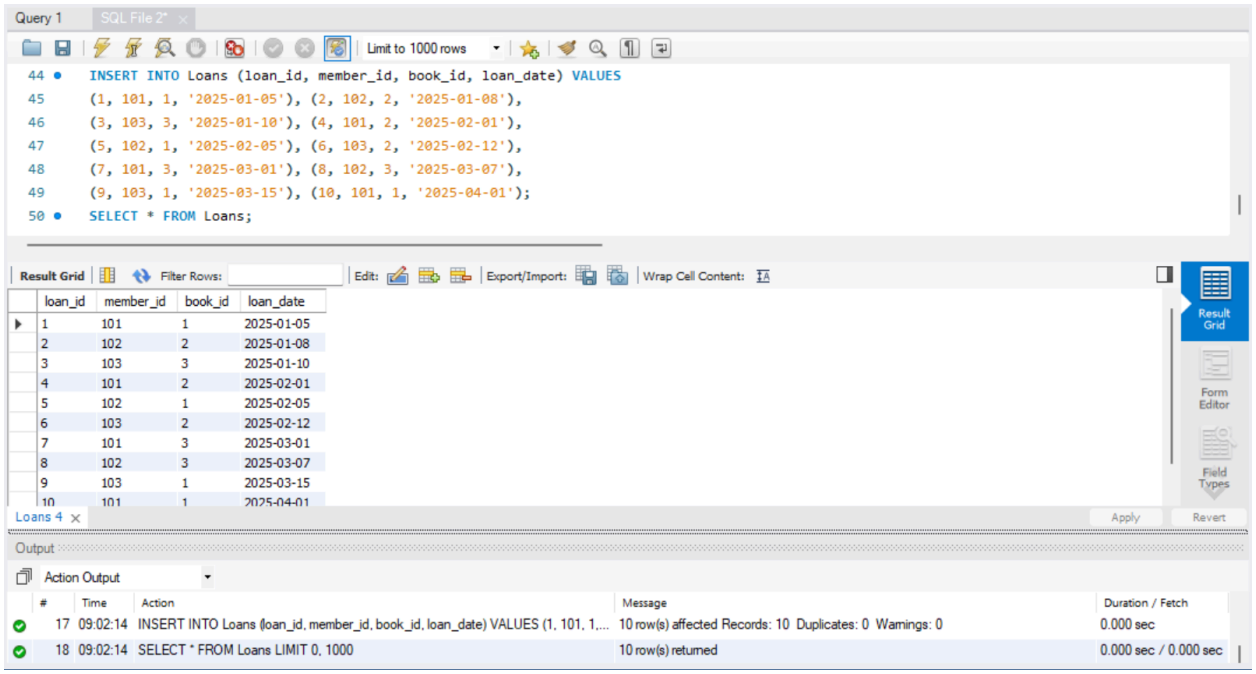
Step 5, 6 & 7 — Members Table: Create, Insert, Display



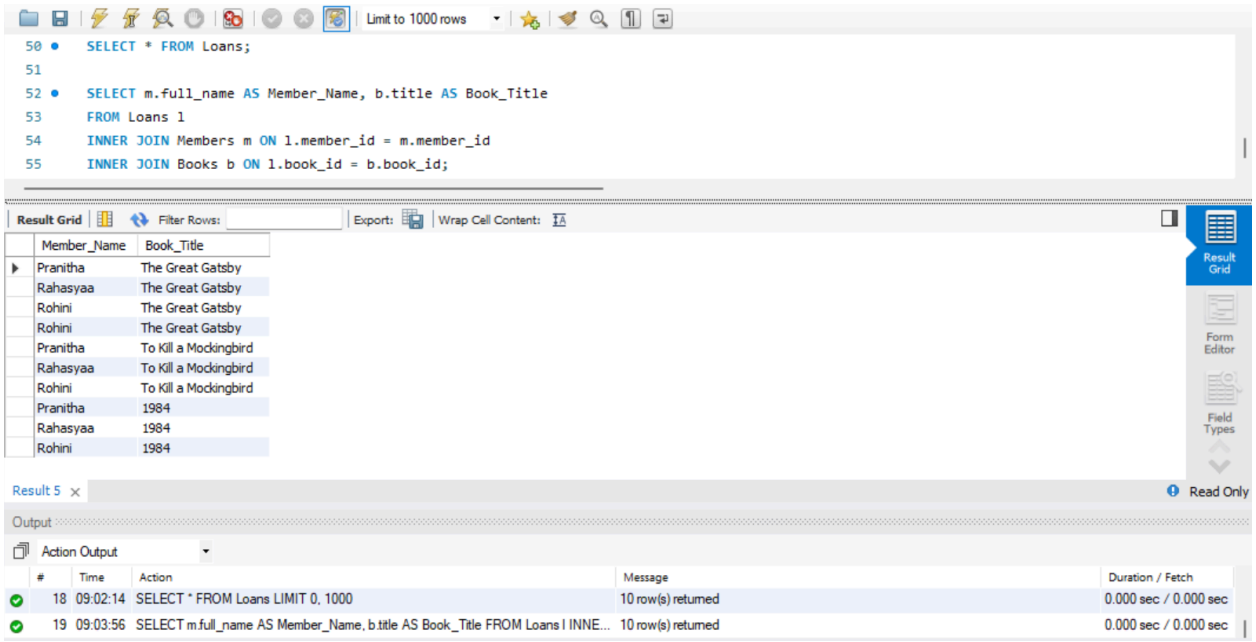
Step 8 — Create Loans Table (FOREIGN KEYS



Step 9 & 10 — Insert 10 Loans and Display



Step 11 — JOIN Query (Who Borrowed What)



Step 12 — GROUP BY Query (Books per Year)

```
56
57 •  SELECT published_year, COUNT(book_id) AS Total_Books
58    FROM Books
59    GROUP BY published_year
60    ORDER BY published_year;
61
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

published_year	Total_Books
1925	1
1949	1
1960	1

Result 6 | Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
19	09:03:56	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNE...	10 row(s) returned	0.000 sec / 0.000 sec
20	09:05:08	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP...	3 row(s) returned	0.000 sec / 0.000 sec

Step 13–16 — Donation via TRANSACTION (Atomicity)

```
61
62 •  CREATE TABLE Donation_History (
63     donation_id INT PRIMARY KEY,
64     book_id INT,
65     donor_name VARCHAR(100),
66     donation_date DATE,
67     FOREIGN KEY (book_id) REFERENCES Books(book_id)
68    );
69 •  START TRANSACTION;
70 •  INSERT INTO Books (book_id, title, isbn, published_year)
71    VALUES (4, 'Animal Farm', '9780451526342', 1945);
72 •  INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
73    VALUES (1, 4, 'Raj Kumar', CURDATE());
74 •  COMMIT;
```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
25	09:06:29	INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_dat...	1 row(s) affected	0.000 sec
26	09:06:29	COMMIT	0 row(s) affected	0.000 sec

Step 17 & 18 — Index on ISBN + Fast Search

```
76 •  CREATE INDEX idx_books_isbn ON Books(isbn);
77 •  SELECT * FROM Books WHERE isbn = '9780451524935';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

book_id	title	isbn	published_year
3	1984	9780451524935	1949
NULL	NULL	NULL	NULL

Result Grid

Books 7 ×

Apply

Output

Action Output

#	Time	Action	Message	Duration / Fetch
27	09:07:38	CREATE INDEX idx_books_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.047 sec
28	09:07:38	SELECT * FROM Books WHERE isbn = '9780451524935' LIMIT 0, 1000	1 row(s) returned	0.000 sec / 0.000 sec